

STAT536HW5

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Problem 1

1.

```
fit_ind <- glm(Freq ~ X1 + X2, family = poisson, data = data)

print(summary(fit_ind))

##
## Call:
## glm(formula = Freq ~ X1 + X2, family = poisson, data = data)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  7.73470    0.01991  388.56  <2e-16 ***
## X1Yes        -0.41638    0.02899  -14.37  <2e-16 ***
## X2Bb_bb      -1.17130    0.03338  -35.09  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 1974.50  on 3  degrees of freedom
## Residual deviance:  312.48  on 1  degrees of freedom
## AIC: 353.17
##
## Number of Fisher Scoring iterations: 4

G2 <- deviance(fit_ind)
df <- fit_ind$df.residual
pvl <- 1 - pchisq(G2, df)

cat("G^2 =", G2, "    df =", df, "    p-value =", pvl, "\n")

## G^2 = 312.4785    df = 1    p-value = 0

fit_sat <- glm(Freq ~ X1 * X2, family = poisson, data = data)

print(anova(fit_ind, fit_sat, test = "Chisq"))
```

```
## Analysis of Deviance Table
##
## Model 1: Freq ~ X1 + X2
## Model 2: Freq ~ X1 * X2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         1      312.48
## 2         0         0.00  1   312.48 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The log-linear model without an interaction term fits the data very poorly. The deviance of the independence model is $G^2 = 312.48$, giving a p-value essentially equal to zero. Therefore, the hypothesis that disease status and SNP2 genotype are independent is rejected. This indicates a strong association between Crohn's disease and the SNP2 genotype.

2.

```
cases <- c(
  data$Freq[data$X1=="Yes" & data$X2=="BB"],
  data$Freq[data$X1=="Yes" & data$X2=="Bb_bb"]
)
controls <- c(
  data$Freq[data$X1=="No" & data$X2=="BB"],
  data$Freq[data$X1=="No" & data$X2=="Bb_bb"]
)

X2 <- factor(c("BB", "Bb_bb"))

logit_fit <- glm(cbind(cases, controls) ~ X2, family = binomial)

print(summary(logit_fit))

##
## Call:
## glm(formula = cbind(cases, controls) ~ X2, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.14787    0.03256  -4.542 5.58e-06 ***
## X2Bb_bb      -1.33248    0.08180 -16.290 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance:  3.1248e+02  on 1  degrees of freedom
## Residual deviance: -1.2568e-13  on 0  degrees of freedom
## AIC: 19.706
##
## Number of Fisher Scoring iterations: 2
```

```
print(exp(coef(logit_fit)))
```

```
## (Intercept)      X2Bb_bb  
##    0.8625430    0.2638215
```

```
print(exp(confint(logit_fit)))
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %    97.5 %  
## (Intercept) 0.8091752 0.9193383  
## X2Bb_bb     0.2242983 0.3091287
```

The logistic model shows that the SNP2 genotype has a very large effect on disease odds. The estimated odds ratio for “Bb or bb” vs. “BB” is $OR \approx 0.26$, with a 95% CI entirely below 1. This means that individuals carrying the minor allele b have disease odds that are only about one-quarter of those with genotype BB. Thus, the minor allele appears to have a protective effect against Crohn’s disease.

3.

```
mat <- matrix(  
  data$Freq,  
  nrow = 2, byrow = FALSE,  
  dimnames = list(  
    X1 = c("No", "Yes"),  
    X2 = c("BB", "Bb_bb")  
  )  
)  
  
print(fisher.test(mat))
```

```
##  
## Fisher's Exact Test for Count Data  
##  
## data:  mat  
## p-value < 2.2e-16  
## alternative hypothesis: true odds ratio is not equal to 1  
## 95 percent confidence interval:  
##  0.2237137 0.3103244  
## sample estimates:  
## odds ratio  
##  0.2638874
```

```
print(fisher.test(mat, alternative = "less"))
```

```
##  
## Fisher's Exact Test for Count Data  
##  
## data:  mat
```

```
## p-value < 2.2e-16
## alternative hypothesis: true odds ratio is less than 1
## 95 percent confidence interval:
## 0.0000000 0.3026314
## sample estimates:
## odds ratio
## 0.2638874
```

```
print(fisher.test(mat, alternative = "greater"))
```

```
##
## Fisher's Exact Test for Count Data
##
## data:  mat
## p-value = 1
## alternative hypothesis: true odds ratio is greater than 1
## 95 percent confidence interval:
## 0.2296763      Inf
## sample estimates:
## odds ratio
## 0.2638874
```

The two-sided Fisher test gives a p-value $< 2 \times 10^{-16}$, confirming a highly significant association. The one-sided test for $OR < 1$ also yields an extremely small p-value, supporting the conclusion that the minor allele reduces disease risk. Conversely, the one-sided test for $OR > 1$ gives $p = 1$, providing no evidence that the minor allele increases risk. Overall, Fisher's tests strongly support that the minor allele b is protective rather than harmful.

Problem 2

1.

```
data <- matrix(c(2037, 631, 327,
                 1757, 18, 200),
              nrow = 2, byrow = TRUE)

dimnames(data) <- list(
  Disease = c("No", "Yes"),
  SNP2    = c("BB", "Bb", "bb")
)

data_df <- as.data.frame(as.table(data))
names(data_df) <- c("X1", "X2", "Freq")
data_df
```

```
##   X1 X2 Freq
## 1 No BB 2037
## 2 Yes BB 1757
## 3 No Bb  631
## 4 Yes Bb   18
## 5 No bb  327
## 6 Yes bb  200
```

```

fit_ind <- glm(Freq ~ X1 + X2, family = poisson, data = data_df)
fit_sat <- glm(Freq ~ X1 * X2, family = poisson, data = data_df)

summary(fit_ind)

##
## Call:
## glm(formula = Freq ~ X1 + X2, family = poisson, data = data_df)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  7.73470    0.01991  388.56  <2e-16 ***
## X1Yes        -0.41638    0.02899  -14.36  <2e-16 ***
## X2Bb         -1.76574    0.04248  -41.57  <2e-16 ***
## X2bb         -1.97398    0.04649  -42.46  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 4650.65  on 5  degrees of freedom
## Residual deviance:  575.91  on 2  degrees of freedom
## AIC: 630.46
##
## Number of Fisher Scoring iterations: 5

anova(fit_ind, fit_sat, test = "Chisq")

```

```

## Analysis of Deviance Table
##
## Model 1: Freq ~ X1 + X2
## Model 2: Freq ~ X1 * X2
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         2     575.91
## 2         0         0.00  2   575.91 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The log-linear independence model fits the data very poorly. The likelihood-ratio chi-square statistic is about 575.9, and the corresponding asymptotic p-value is effectively zero. This means that the hypothesis of independence between Crohn's disease status and the SNP2 genotype is rejected. There is very strong evidence of an association between the two variables.

2.

```

fisher_test <- fisher.test(data)
fisher_test

##
## Fisher's Exact Test for Count Data

```

```
##
## data: data
## p-value < 2.2e-16
## alternative hypothesis: two.sided
```

```
fisher_test$p.value
```

```
## [1] 1.920788e-125
```

The generalized Fisher's exact test for the 2 by 3 table gives an exact p-value of approximately 1.9 times 10 to the power minus 125, which is essentially zero. This confirms the same conclusion as the asymptotic test: there is overwhelming evidence that disease status and SNP2 genotype are not independent.

3.

```
cases <- c(
  data["Yes", "BB"],
  data["Yes", "Bb"],
  data["Yes", "bb"]
)

controls <- c(
  data["No", "BB"],
  data["No", "Bb"],
  data["No", "bb"]
)

X2 <- factor(c("BB", "Bb", "bb"))

logit_fit <- glm(cbind(cases, controls) ~ X2,
  family = binomial)

summary(logit_fit)

##
## Call:
## glm(formula = cbind(cases, controls) ~ X2, family = binomial)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.49164    0.08977  -5.477 4.33e-08 ***
## X2Bb         -3.06529    0.25534 -12.005 < 2e-16 ***
## X2BB          0.34377    0.09549   3.600 0.000318 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5.7591e+02  on 2  degrees of freedom
## Residual deviance: 6.4837e-14  on 0  degrees of freedom
## AIC: 26.057
```

```
##  
## Number of Fisher Scoring iterations: 3
```

```
exp(coef(logit_fit))
```

```
## (Intercept)      X2Bb      X2BB  
## 0.61162080 0.04664025 1.41025773
```

```
exp(confint(logit_fit))
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %    97.5 %  
## (Intercept) 0.51224862 0.72845249  
## X2Bb        0.02735247 0.07484574  
## X2BB        1.17071475 1.70254159
```

The logistic regression model shows that the SNP2 genotype strongly affects the probability of having Crohn's disease. The estimated odds ratio for the Bb genotype relative to BB is about 0.047, indicating that Bb is associated with greatly reduced disease odds. The odds ratio for the bb genotype relative to BB is about 1.41, indicating moderately increased disease odds. Overall, the regression results show that Bb appears protective, while bb increases the odds of disease compared to the BB genotype.